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Latin *crassus*, *grossus*, *classis*

Phonology and etymology

Abstract: According to Christol (1996: 809ff.), some Latin words in *-ss-* (allegedly with "expressive gemination") could receive a principled explanation as systematic exceptions to rhotacism. At the time of rhotacism, conservative dialects might have remained at the stage with intervocalic /-z-/ (as in Oscan). Then, speakers in rhotacizing dialects (or using a rhotacizing phonostyle) would identify the intervocalic [z] with [ss], at that time the only intervocalic sibilant in rhotacizing speech; cf. Lat. ⟨ss⟩ rendering Gk. ⟨ζ⟩ (= [z]). Christol's own examples are not strongly supportive; but Lat. *crassus* 'thick, fat', *grossus* 'unripe fig' (perhaps also *grossus* 'thick, coarse'), and *classis* 'levy' could provide good examples of the process. The etymologically obscure *crassus* and *grossus* can thus be given satisfying accounts based on attested PIE *s*-stems, *s*-presents, and/or *s*-enlarged forms. (The discussion of *crassus* includes analysis of the neglected form *crassundia* 'sausages', which may support an old *s*-present.) For *classis*: the root etymology was clear (cf. Lat. *calāre* 'call, summon', U. **kařetu**/*carsitu* 'he shall call', Gk. καλέω 'call', etc.), but the *-ss-* formation was not; it can now be interpreted, given Christol's theory, via *s*-forms attested within Latin (e. g. *Calābra*) and elsewhere (cf. the *s*-present in Hitt. *kalliš-*^{zi} 'call, summon').

Keywords: Latin etymology, rhotacism, exceptions to rhotacism, expressive gemination, *s*-stems, *s*-enlargements, *s*-presents, *sermo rusticus*, *sermo castrensis*

1 Lat. *crassus* 'thick, fat'

1.1 Lat. *crassus* 'thick, dense, solid, fat; crude' (Pl.+) has no good etymology. Traditionally,[1] the word is compared with Lat. *crātis* 'wickerwork; fence' (Pl.+), which belongs with Baltic and Germanic words for 'fence; door' (such as Eng. *hurdle*). Yet this material is itself problematic and ultimately obscure: the formal mismatches among the alleged cognates, as well as their semantic specificity (DELL: "[t]erme technique"), may point to a non-IE source, as de Vaan suspects,

1 E. g. WH: s. v.; IEW: 584.

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perhaps correctly (EDL: s. v. *crātis*).[2] Thus the comparison between *crassus* and *crātis* is rightly rejected by DELL and EDL (all s. v. *crassus*).

It is surely significant somehow (as often noted) that *crassus* is one of many words with medial geminate describing physical deformities or other negative features: so also, among others, *brocc(h)us* 'buck-toothed', *cloppus* 'deformed', *gibber* 'hunchbacked' (cf. *gibbus* 'hump'), *maccus* 'buffoon', including some used as cognomina: besides *Crassus* 'Fatso' itself, e. g. *Bassus* 'id.',[3] *Flaccus* 'Lop-Ears'.[4] It is worth recalling, however, that some forms usually placed in this category may have medial gemination resulting from bona fide phonological processes (rather than "expressive gemination"). Thus, for example, *lippus* 'bleary-eyed' (Pl.+) may display a Sabellic labial, with gemination resulting from application of the "*littera*-Gesetz" to an OLat. **leipos* (Fortson 2008).[5]

Most recently, de Vaan (in his EDL entry) has offered the following comment, based on the geminate in *crassus*: "it is conceivable that the older form was **crāsus*. This, however, does not clarify the etymology." But I suggest that this insight may well provide the best approach to the etymology of *crassus*.

1.2 If the starting point was a (pre-Lat.) **krāso-*, one might consider first whether rhotacism in such a form could have been blocked by the presence of /r/ in the onset cluster: cf. the well-known failure of rhotacism in forms like *miser* 'wretched' (fem. *misera*, etc.), *rosa* 'rose', *caesariēs* 'long hair' (both Pl.+), where the blockage is said to involve *s* ... *r* (or *r* ... *s*) in consecutive syllables.[6] This is possible, and I return to this point below; but even in that case, it would be necessary to explain why (apart from vague appeals to "expressive gemination") the unrhotacized outcome would be /krasso-/ (or /krāsso-/, at least at first). There is in fact a plausible theory

2 Garnier (2010: 167) attempts to etymologize *crātis* as a *ti*-stem abstract **ḱr̥h₁-tí-* (to the root of Lat. *crēscere* 'grow', on which see below); this is formally impeccable but semantically far-fetched. See further below (1.6 ad fin.) on Garnier's approach to *crassus*.

3 More on *bassus/Bass(i)us* in 3.2 (1.) below.

4 See on such forms Leumann (1977: 182, with explicit reference to "expressive gemination"; similarly Grammont 1933: 382) and van Ooteghem 1963: 65 n. 6 (on the onomastic use).

5 On the typology of gemination (with some attention to "expressive gemination" or "affective lengthening"), see Blevins 2008 and Dmitrieva 2012.

6 On exceptions to rhotacism (and for a phonological analysis of rhotacism more generally), see recently Roberts 2012 (and specifically 2012: 81, 88f., 91 on arguments for a constraint against **rV̆r* sequences). For loans of the type *cisium* 'two-wheeled carriage', *basis* 'base', *asinus* 'ass', see Weiss 2011: 151 ("They were not in Latin at the time of Rhotacism"); and other exceptions involve etymologically difficult (and phonologically ambiguous) material, e. g. *casa* 'cottage' (Ter.+; see EDL: s. v. for various possibilities). On the phonetics and acoustics of rhotacism (including Latin): Painter 2011. (Neither Roberts nor Painter refers to the *-ss-* cases under discussion here.)

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that may account for this development, suggested by Christol,[7] whether or not his own examples (which do not include *crassus*) are actually supportive. Christol proposes that at the time of rhotacism, conservative dialects might have remained for some period of time at the stage with intervocalic /-z-/ (as in Oscan: gen. pl. f. *egmazum* 'rerum' [cf. Lat. *-ārum*], etc.). Then, speakers in fully-rhotacizing dialects (or preferring to use a rhotacizing phonostyle) would identify the [z] of /VzV/ forms with [ss], at that time the only intervocalic sibilant in rhotacizing speech; cf. Lat. ⟨ss⟩ rendering Gk. ⟨ζ⟩, phonetically [z] in Hellenistic times.[8] For original /V:zV/ sequences, at first heard as /V:ssV/ by rhotacizing speakers: once the Classical Latin *ss*-simplification had occurred (OLat. *cāssus, caussa* > Class. *cāsus, causa*, etc.) — i. e., with /V:ssV/ no longer a permitted sequence[9] — such unrhotacized forms could have entered the speech of rhotacizing speakers in the form /VssV/ or (conceivably) /V:sV/.[10] If the conservative source-dialect in question was rural (or otherwise non-standard), such borrowed unrhotacized *ss*-forms might have had a rustic or expressive ring, or could have been used in situations where a high-style alternative did not exist.

Christol's examples, unfortunately, are not clear-cut:

1. *cassus* 'empty, lacking' (Pl.+): but the rusticity or expressivity of this word is questionable, and there is in any case no independent evidence for a thematic basis **kas-o-* or **kās-o-* (or, if one prefers, **k(e)h*$_2$*s-o-*).[11] Perfectly plausible, instead, would be a thematized *s*-stem **kas-(e)s-o-* beside the *ē*-stative **kas-ē-* (Lat. *caret* 'is lacking' and Fal. **carefo** 'I will lack', O. **kasit** 'it is necessary'), cf. *egestās* 'poverty', *egēnus* 'lacking' (< **eges-no-*) beside *egēre* 'be in need'.
2. *assus* 'roasted, grilled, dry' (Pl.+; ASOM CIL: I², 560 [3rd c.]): but there are many possibilities besides **āso-* (~ *ārēre* 'be dry'), including **ars(s)us* (cf. *ardēre*

7 Christol 1996: 809ff. cf. Weiss 2011: 151 n. 14.

8 On Gk. medial -ζ- ~ Lat. ⟨ss⟩, see Biville 1990: 108ff. (112ff. for Gk. -ιζω and Lat. *-issō*).

9 With the restricted exception of verbal forms of the type *audīsse* (< *audīvisse*) and (analogically) *amāsse* etc. (Leumann 1977: 181) and, at least according to some theories, the type *amāssō, prohibēssīs*, etc. (Leumann 1977 loc. cit.; de Melo 2007: 315ff.); perhaps also (via Lachmann's Law) an original **pēssimus*, later subject to analogical retention of *-ss-* (based on the geminate in *-issimus* superlatives) and shortening to *pessimus* (Jasanoff 2004: 412 n. 13).

10 As pointed out to me by Dieter Gunkel, the choice of /VssV/ (over /V:sV/) could have been supported by influence from other geminated forms that were interpretable as "expressive" or "affective" in some way; cf. above on *brocc(h)us, cloppus*, etc., and (for analogical retention of geminate *-ss-*) cf. also *pessimus* (preceding fn.).

11 On the reconstruction of the root, see Garnier 2013: 64 with n. 39, arguing for alignment with **ḱes-* 'cut', whence resyllabified **ḱas-ó-* (presumed on the basis of *cassus* itself, which he takes to show expressive gemination, citing Christol); **ḱas-ó-* is then taken as the source of a denominative stative **ḱas-eh*$_1$*-(ye/o-)*.

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'burn'), **azd-to-* (cf. ἄζω 'dry'), and **ad-to-* (cf. *ador* '[dried] grain'), among others (see EDL: s. v.).

3. *nassus, nassum* 'nose' (Naev.+), cognomen *Nassō*: traditionally spelled *nāsus/-m, Nāsō* (cf. *nārēs* 'nostrils'); but MSS often have *nass-* (e. g. Pl. *Mer.* 310 *nassum* [*A*]), and inscriptional NASS- is also found (e. g. NASSO CIL: IV, 3204). This may be a relatively good case (cf. above on /V:sV/ ~ /VssV/), though a derivative from a secondary *s*-stem **nas-s-* may also be possible (as Weiss points out, 2010b).

In short: Christol's theory is attractive, but lacks support in the form of clear examples.[12] I suggest, however, that pre-Lat. **krāso-* > Lat. *crassus* may constitute one such good example. Specifically, I propose that **krāso-* belongs with the family (a term used loosely for the moment) of Lat. *crēscere* 'grow', *creāre* 'make grow, create', *crēber* 'thick, dense, frequent', *incrēmentum* 'growth, increase', *prōcērus* 'tall', *Cerēs* (growth goddess) = O. (dat.) **kerrí**, among other forms often placed together with these.[13]

1.3 It has been questioned whether all of these forms (and others) actually belong together, an uncertainty related to the problem of how many roots to reconstruct for such forms, and with what shapes. Broadly speaking, there are three main positions in the recent literature:

1. A root **ḱreh$_1$-*/**ḱerh$_1$-* 'grow' (with Schwebeablaut) underlies forms such as Lat. *crēscere, prōcērus*, HLuv. **zīra-* in *zīralamma/i-* 'fruitful', but is distinct from **ḱerh$_3$-* 'feed' (Gk. [aor.] ἐκόρεσα 'satiated', Li. *šérti* 'feed', etc.).[14]
2. Everything belongs to **ḱerh$_3$-* 'feed': thus e. g. *crēscere* would derive from a Narten *s*-present **ḱrēh$_3$-s-*; for the semantics, cf. Lat. *alere*, OIr. *ailid* 'feed, nourish' vs. Go. *alan* 'grow'.[15]

12 Dieter Gunkel points out to me that Meiser's account of the maintenance of unrhotacized *-ss-* in Latin superlatives in *-issimus* (Meiser 1998: 153) has certain affinities with Christol's theory; but the problem of the gemination in Lat. *-issimus/-errimus/-illimus* is more complex, as Gunkel has discussed (2012). It should be noted, parenthetically, that some cases of Lat. surface *-s(s)-* beside *-r-* (from rhotacism) have nothing to do with the phenomenon in question: e. g. Lat. *quaesō* 'seek' (inscriptionally with ⟨-ss-⟩) beside *quaerō* 'ask' is surely an old desiderative, with historical *s*-suffix; similarly the perfect of *quaerō* itself (*quaesīvī*), based on an *s*-aorist.

13 Recent additions: OIr. *forcraid* 'excess', MW *gwargred/gworgred* 'surplus' (Weiss 2013: 351).

14 Rieken 2003; García Ramón 2010; LIV² Add.: s. v. **ḱreh$_1$-*; as a variant: Garnier (2010: 165–169) aligns ἐκόρεσα (and other Greek material) with Lat. *crēscere* etc.

15 LIV²: s. v. **ḱerh$_3$-*; Villanueva Svensson 2012. Similarly Viredaz 2005, but with a twist for *crēscō*: actually < **g^{h}reh$_1$-* 'grow' (Eng. *grow* etc.; more below on this root), whence Ital. **χrē-*, remodeled to **krē-* by contamination with Italic **ḱerh$_3$-* forms.

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3. Lat. *crēscere* (and other Italic material) belongs rather with *$k^{(w)}reh_1$- 'become stronger', cf. Slav. **krějǫ*, **krĭjati*/**krějati* 'recover one's strength, heal' (SCS *o-krijati*, Russ. dial. *kreját'*, etc.).[16]

The details of this controversy can be left aside here, since pre-Lat. **krāso-* would derive from a zero-grade "**k̥rH-s-o-*", a form that could in principle be aligned with any of the roots in question and which itself would shed no light on the larger problem. Rather, the main issue here is to justify such a form both morphologically (i. e., by defending a zero-grade form with an *s*-"extension" of some kind) and semantically (i. e., by positing a semantically credible source for a word that means 'thick, fat', although this point is almost settled by the meaning of Lat. *crēber*).

As for the morphology, evidence for zero-grade forms belonging to any of the roots in question is not abundant, and there is still less evidence for *s*-extended forms belonging to such zero grades; but there is some, which can be considered in terms of two approaches: a noun-based analysis (i. e., based on an *s*-stem; 1.4), and a verb-based one (i. e., based on an *s*-present, or else a verbal root with *s*-enlargement; 1.5).

1.4 One starting point for a thematic **krā-s-o-* would be an *s*-stem, which certainly exists: within Italic itself, cf. Lat. *Cerēs* (1.2), pointing to a hysterokinetic *$^{(o)}$*ḱerH-és*,[17] and Italic thematized derivatives of the types **keres-o-* (e. g. Lat. *Cer(r)us Manus* 'creator bonus' [Carm. Sal.], U. dat. **çerfe**) and **keres-yo-* (e. g. U. dat. sg. f. **çerfie**). But there are also traces of an old *s*-stem elsewhere, as in OHG *hirso* 'Hirse' < Gmc. **her-s-ija(n)-*, together with some less perspicuous Armenian data; and Stüber (2002: 117) has plausibly argued that this material, together with hysterokinetic *$^{(o)}$*ḱerH-és*, permits the reconstruction of a proterokinetic *s*-stem **ḱérH-os*. Indeed, the existence of such an *s*-stem underlying *Cerēs* was already posited by Schindler (1975: 63), who later[18] added Hitt. (neut.) *karaš* (i. e. [kars]) '(type of) wheat', with zero grade of the suffix.[19]

16 Hill 2006, apparently followed by de Vaan (2012: 319, on the morphology of *creāre*). But see LIV² Add.: s. v. **ḱreh$_1$*- on Slavic *kr-* < **ḱr-*.

17 As observed by Hill (2006: 198 n. 14), *Cerēs* could be decompositional to an original compound adjective: e. g. **h$_1$su-ḱerH-és* 'mit gutem Getreide' → **ḱerHés* 'die Getreidige'. (For other analytical possibilities, see Schindler 1975: 63f., and see further below, n. 27.)

18 Class instruction, ca. 1976.

19 On Hitt. *karaš*, see Melchert 2013: 181, including arguments (181 n. 19) against the popular interpretation (see EDHIL: s. v., with references) in terms of a root noun **ǵ^h(e)rs-*, related to Lat. *hordeum*, OHG *gersta* 'barley'. On the Armenian forms (*ser/seṙ* 'offspring', *sermn* 'seed', etc.), see especially Olsen 1999: 76, 85f.; Viredaz 2005: 154ff.; and Hill 2006: 200ff. There is also some Italic

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Needed next is a thematic derivative of such an *s*-stem, for which the most familiar type is the exocentric possessive derivative routinely exemplified by Ved. *vatsá-* 'calf, yearling', i. e. **wet-s-ó-* 'having a year' ← **wét-es-* 'year'.[20] Less well known is the fact that such thematic derivatives can appear "zeroed-out" (i. e., with "double zero grade"), as in Ved. *śūṣá-* 'powerful' < **ḱuH-s-ó-*, cf. the *s*-stem basis in *śávas-* 'power' (Rau 2009: 128).[21] And there are at least two candidates within Latin for such double-zero-grade thematic derivatives of *s*-stems:

1. *russus* 'red' (Enn.+) < **h₁rudʰ-s-o-*, cf. primary *s*-stems in Lat. *rubor*, Gk. ἔρευθος 'redness';[22]
2. *saxum* 'rock' (Pl.+), as if **sₑk(H)-s-o-*, to **sekH-* 'cut' (Nikolaev 2010: 231), cf. Gmc. **sah-s-a-* 'sword' (with secondary *o*-grade: cf. Gmc. 'saw' < **sok(H)-eh₂*, 'sedge' < **sok(H)-yo-*).[23]

All of the necessary ingredients, then, are available for interpreting a pre-Lat. **krāso-* in terms of an old **ḱr̥H-s-ó-* 'fat' (*'having growth'), a thematic adjectival derivative (with double zero grade) based on an original *s*-stem **ḱérH-es-* 'growth' (or 'nourishment'), indirectly attested within Latin itself via its hysterokinetic internal derivative *Cerēs* and thematic derivatives based on it in turn.

1.5 One can also consider the problem from the perspective of verbal morphology. As already mentioned (1.3 (2.)), Villanueva Svensson (2012) has proposed that all of the 'grow'/'nourish' material belongs to the root **ḱerh₃-* 'feed', which in his analysis made an original Narten *s*-present (or "*s*-desiderative") **ḱréh₃-s-*/**ḱréh₃-s-*. Thus

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data that could conceivably be derived from **ker-s-* (as opposed to **ker-es-*), such as Lat. *cēna*, O. **kersnu** 'meal' (Schrijver 1991: 432); but these forms have other possible etymological sources, on which see EDL: s. v. *cēna*.

20 Elsewhere (Vine 2009: 216) I propose a somewhat "deeper" analysis of the *s*-stem traditionally seen here (and therefore of this thematic derivative); but this does not affect the point at issue, for which other examples could easily be supplied: see e. g. Stüber (2002: 31) on **h₃ór-es-* 'rise' (Gk. ὄρος 'mountain') → **h₃or-s-o-* 'rump' (Gk. ὄρρος etc.).

21 For 'year', in the traditional analysis as an *s*-stem, note **ut-s-o-* in CLuv. *uššа/i-*, HLuv. *u-sa/i-*, Lyc. *uhe/i-* 'year'. Similarly, at least traditionally, Ved. *útsa-* 'well, spring' (YAv. ⁺*usa-*) from an *s*-stem based on the word for 'water'; but see again Vine 2009: 220 for a different conception.

22 Differently Hill (2003: 224f.): *russus* < **h₁rudʰ-to-*, based on the root noun in OIr. *rú* 'red color' and other forms. But such a *to*-formation is difficult to parallel in Latin, and the root noun is not attested in Italic. Hill's analysis, in any case, was predicated on the incorrect assumption that an *s*-stem derivative must have full grade of the root (thus here **h₁reudʰ-s-o-* > Lat. ˣ*rūso-*). A thematic **h₁rudʰ-s-o-*, moreover, provides a good basis for the substantivization **h₁rudʰ-s-i-* 'redness' that may be seen in OIr. *ruis* 'holly' (Weiss 2013: 345 with n. 56).

23 On *russus* and *saxum* in these terms (with additional examples of this type), see recently Höfler 2015a: 35 and (more generally on the type) 2015b.

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Lat. *crēscere* would only have become a *scō*-verb secondarily, like *pāscere* 'feed' (PIE **peh*$_2$*-s-*: cf. Hitt. *paḫs-* 'protect', OCS *pasǫ* 'graze'). According to Villanueva Svensson, this conception is further supported by Lat. *crēber* (1.2) < pre-Lat. **krē-s-ro-* (as opposed to the traditional interpretation **krē-θro-*, with a PIE suffix **-d*^{h}*ro-* — appropriate for instrument nouns, but difficult to justify in such an adjective).[24]

Although not noted by Villanueva Svensson, additional support for his conception might come from Lat. *crassundia* (neut. pl.) 'thick (i. e. stuffed) intestines' (i. e. sausages) (Varro *L.* 5.111). The form is generally ignored, except to make one of the following assertions:

1. it has nothing to do with Lat. *-undus* forms (so Leumann 1977: 331, without explanation);
2. it is modeled on *crepundia* (WH, following Muller (1926: 110) and others; similarly OLD: s. v.);
3. "a *crassus* ductum esse videtur" (TLL: s. v. [Thurneysen]; cf. the similarly vacuous OLD characterization "*crassus* + *-undia*").

Formally, the sole Latin comparandum to *crassundia* is indeed the neut. pl. *crepundia* (Pl.+), just mentioned; but the form is controversial, beginning with its meaning. The standard definition 'baby's rattle' (TLL; OLD; EDL) suggests comparison with *crepitāculum*, *crepitācillum* 'id.' (Lucr.+) and ultimately *crepō* 'make a sharp noise' (Pl.+). But in Plautus, the word refers to trinkets worn *around a child's neck*, sometimes functioning as ἀναγνωρίσματα 'recognition tokens' — cf. the secondary sense 'swaddling blanket' (Plin.+), clearly based on the recognition usage; and clear attestations with a meaning 'noisemaker' are attested only late (Valerius Maximus, Apuleius, Prudentius, Justinian). Thus, according to Leumann (1933: 240ff.), the word originally had nothing to do with *crepō*, but is rather related to the *bulla aurea* (a ceremonial pouch worn around the neck, allegedly of Etruscan origin), and therefore is probably based on an Etruscan word. This seems hard to rule out, and may be correct; nevertheless, Meillet (in DELL: s. v.) still allows for *crepundia* as a derivative of *crepō*, and this may also be correct. If, then, *crepundia* is native: despite *crepāre* (perf. *crepuī*, *crepitum*), an original participial **crepundus* 'noise-making' (whence a substantivization **crepundium*, pl. *-ia*) could

24 Viredaz (2005: 163 n. 108) attempts — in my view unsuccessfully — to dissociate the alleged "**-d*^{h}*-ro-*" here from instrument nouns in **-d*^{h}*ro-*; but he also considers **krē-s-ro-* to be a possibility. Lat. *crēber* is analyzed in terms of a pre-Lat. **krē-s-ro-* by Hill (2006: 195f.), though with a different overall conception (cf. 1.3 (3.)). Far less attractive is Garnier's hypothetical basis **ḱ̥rh*$_1$*-d*^{h}*(h*$_1$*)-ró-* (2010: 167), with unsupported formant **-d*^{h}*(h*$_1$*)-* (see 1.6 below, ad fin.) and requiring the vocalism of *crēber* to have been remade after *crēscō*.

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have been based on the thematic present (*)*crepere* that is often assumed and perhaps directly attested in 2 sg. *percrepis* (Varro *Men.* 124 = Non. p. 255M, 387L).[25]

The more important question here is whether Lat. *crassus* could have been modeled on *crepundia* so as to generate the form *crassundia* (cf. the above assertions, especially 2. and 3.). This seems unlikely in the extreme: the words have little in common apart from their onset cluster [kr-], surely an insufficient basis for such an analogical formation. If anything, a theoretical **crassundus* 'growing fat' (vel sim.) might have a greater claim to antiquity than **crepundus*: such a form could have been supported by *rotundus* (a member of the original set of core *-undus* adjectives) at the (quite early) stage when it had already developed the meaning 'round' (Cato+). The most likely basis for a form **crassundus*, however, would not have been the already-existing adjective *crassus*, but rather a verb **crassor* 'grow fat' (cf. **retor* or **rotor* 'roll' → *rotundus*).[26]

Thus, to pursue the *s*-present scenario in more detail, one can posit the following derivational steps and relationships.[27] The originally acrostatic *s*-present **ḱrḗh$_3$-s-*/**ḱréh$_3$-s-* 'grow' was renewed as an ablauting present **ḱrḗh$_3$-s-*/**ḱr̥h$_3$-s-ˊ* with mobile accent.[28] The strong form **ḱrḗh$_3$-s-* ended up as Ital. **krēske/o-* (cf. *pāscō*, as above),[29] while the weak form **ḱr̥h$_3$-s-(énti)* was thematized as **ḱr̥h$_3$-s-e/o-* (cf. OCS *pasǫ*). Thematic **ḱr̥h$_3$-s-e/o-* > pre-Lat. **krās-e/o-*, whence **crassor* 'grow fat' (with verbal adjective **crassundo-* → substantivized *crassundia*, as above). Finally — and most importantly for present purposes — verbal **ḱr̥h$_3$-s-e/o-* could also have produced a deverbal "zero-grade τομός form" **ḱr̥h$_3$-s-o-* 'growing',[30] whence pre-Lat. **krās-o-* and ultimately (by Christol's theory) *crassus* 'fat'.

25 On (*)*crepere*, see Meiser 2003: 137f., 232 (type *sonere* ~ *sonāre* 'sound, make a noise', etc.). The existence of a **crepundus* remains possible even if *crepere* itself is only secondarily thematic; see on this point Bock 2008: 214f., with references.

26 On *rotundus* (originally *'disposed to roll') and its parallel "adjectives of motion" *secundus* 'following', *oriundus* 'rising', *lābundus* 'slipping', see Jasanoff 2006.

27 For expository convenience, I adopt the root shape and morphological assumptions of Villanueva Svensson's analysis, which has the benefit of accounting also for the lengthened grade in Lat. *prō-/sin-cērus* (Villanueva Svensson 2012: 243, following Rieken 2003, who credits Nussbaum for this approach), and even (as the *IF* referee points out to me) the full grade root in hysterokinetic *Cerēs* (i. e., without a need to assume a decompositional origin, cf. n. 17 above); but one can easily substitute, *mutatis mutandis*, other versions of the PIE root, as in 1.3 (1., 3.) above.

28 See Jasanoff 2003: 42 on such zero-grade renewals in the weak stem-form of acrostatic paradigms.

29 With a *ś́ke/o-*present perhaps also attested in the HLuv. participle /zarzamis/ 'heranwachsende' (Rieken 2003).

30 See on such formations Nussbaum 2007a: §7, further 2007b on secondary verbal nouns built to *s*-extended verbs (e. g. Av. *sraoša-* 'obedience', OCS *sluxŭ* 'hearing', cf. thematic verbal **ḱléw-s-e/o-* in Ved. *śroṣa-* etc., LIV²: s. v. **ḱleus-*). As a parallel for a nominal **ḱr̥h$_3$-s-o-* plausibly derived from

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1.6 To conclude the discussion of Lat. *crassus*, we must begin by admitting that each of the options just considered, i. e. pre-Lat. **krā-s-o-* < **ḱr̥H-s-ó-* on the basis of either an *s*-stem (1.4) or an *s*-present or *s*-enlargement (1.5), is uncomfortably hypothetical to a certain extent, the second perhaps more so than the first: an *s*-stem is probably reconstructable for the protolanguage and is attested in Latin, though not in the actual double-zero-grade version required; and Lat. *crēscō*, like *pāscō*, could in theory be based on an *s*-present, possibly supported by *crēber* and *crassundia*, but not directly attested. Nevertheless, it can also be said, first of all, that the morphological specificity of these analyses renders both proposals superior to mere root etymologies; further, that a basis in forms meaning 'growth', 'nourishment' or (in the verbal analysis) 'grow' or 'feed' provides a semantically attractive source for an adjective meaning 'thick, fat'; and finally that the phonological development is directly accounted for by Christol's theory about non-rhotacism and *s*-gemination (1.2), including the possibility, in this case, that a term such as *crassus* could have been associated with some sort of *sermo rusticus* or lower-register speech form. These suggestions, then, seem preferable to the traditional idea of cognation with Lat. *crātis* (1.1), and, if correct, would provide a good example of the process postulated by Christol.

A final comment concerns Garnier's (2010: 165f.) attempt to etymologize *crassus* in terms of the root of Lat. *crēscere* . Naturally, I am in sympathy with the root connection; and Garnier draws suggestive parallels between *crēscere*, which he takes to continue an intransitive inchoative **ḱr̥h₁-dʰ(h₁)-sḱ-e/o-* (with *ā*-vocalism remade to /ē/ after the aorist), and Lat. *suēscere* 'become accustomed to'. If, as Garnier suggests, *crassus* continues **ḱr̥h₁-dʰ(h₁)-tó-* (as the original verbal adjective in **-tó-*, later supplanted by *crētum*), the outcome should have been **crāssus* (Class. Lat. ˣ*crāsus*), which might also provide an appropriate input for the development envisaged by Christol's theory. But, in distinct contrast with *suēscere* (cf. OLat. SVODALES 'sodales', Ved. *svadhā́* 'autonomous power'), there is little independent evidence for **ḱr(e)h₁ + *dʰ(e)h₁-*, which remains a hypothetical construct.[31] It thus seems preferable to operate with the above analysis in terms of actually-attested *s*-formants.

a verbal *s*-formation, note further (as the *IF* referee points out to me) the Sanskrit *sā*-abstracts to *sa*-desideratives (Ved. *jígīṣa-* 'want to win' → *jigīṣā́-* 'desire for victory' etc., AiG: 2.2, 242f.).

31 Garnier (2010: 166 n. 16) connects Gk. κόρθυς 'heap (of grain)' and Ved. *śárdha-* 'troop', allegedly based on meanings involving 'swelling'; but his "pseudo-root" **ḱerdʰ-* for these words (if that is indeed their source, which is far from clear; see e. g. DELG: s. v. κόρθυς) hardly constitutes real support for an original **ḱr(e)h₁ + *dʰ(e)h₁-*.

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2 Lat. *grossus* 'unripe fig; green, unripe; thick, fat, coarse'

2.1 It is convenient to take de Vaan's lemma (EDL: 273) as a starting point for the discussion of Lat. *grossus*: "**grossus** 'immature fig' [m. *o*] (Cato+); 'thick, unripe' [adj.] (Col.+)". This information can be corrected and supplemented as follows:[32]

1. *grossus* in the meaning 'unripe fig' is both m. (e. g. Matius, Celsus) and f. (Pliny) in literary sources (but see (4.) below);
2. also attested is a diminutive *grossulus* '(small) unripe fig' (Postumus ap. Macr., Col.);
3. for *grossus* as an adjective: despite de Vaan, the word does not mean 'thick' in Columella, but only 'unripe', with reference to grapes and apples; elsewhere also (Schol. Germ. Bas. 100.16) with reference to figs;
4. the above usage of *grossus* as an adjective 'unripe, immature (of fruit)' seems to have led to a late substantivization *grossum* (Hil. in Matth. etc.), or else (as in WH) *grossus* (m., f.) has developed as *grossum* (n.) after *pōmum* '(tree-)fruit';
5. finally, and very importantly, *grossus* in the meaning 'thick, fat, coarse' is restricted to Late Latin (Tertullian, Augustine, Vulgate, Cassiodorus, Sulpicius Severus, late glosses, etc.), perhaps apart from a single appearance in Paulus ex Festo (40.13L, but as part of a gloss, and thus possibly also late).

Thus, although "Latin *grossus*" is often treated as a single word, whether implicitly (as in EDL) or explicitly (WH: *grossus* 'unripe fig' "identisch mit *grossus* 'dick'"), this may not be true in some sense. The bare philological facts suggest that at least descriptively, one must distinguish between two quite different lexical entities:

1. "*grossus*[1]", an agricultural term referring to unripe or immature fruit, especially figs: the noun (m., f., later n.) is clearly old (Cato, Matius) and well-attested in Augustan and early Imperial Latin (Celsus, Pliny, Columella), and the adjective is at least as old as Columella;
2. "*grossus*[2]", the adjective meaning 'thick, fat, coarse': this usage (or this word) may be restricted to Late Latin, and became well-established in Romance (> Fr. *gros*, Sp. *grueso*, etc.; cf. also the adjective **grassus*, from contamination with *crassus*, whence Fr. *gras*, Ital. *grasso* 'fat', etc.).

This is the conception implied by DELL, who divide the material into two separate entries, with an etymological note provided only for "*grossus*[2]".

32 See TLL: s. v. for the immediately following points.

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2.2 Indeed, the etymological literature addresses only *grossus*[2]. The traditional approach[33] sets up a preform **g^{w}retso-*, comparing MIr. *bres* 'great', somehow a variant of OIr. *bras* (MW *bras*, MBr. *braz*) 'id.' and also 'boastful, defiant, violent'. The same account is maintained by Mallory & Adams (2006: 299) and appears in some Celtic lexica (e. g. Deshayes 2003: 132), cf. EDPC: 74: "Although this etymology is disputed, the formal and semantic parallelism between these Celtic words and Lat. *grossus* is too obvious to be dismissed lightly". Vendryes, however, is skeptical (LEIA: B, 79, raising both formal and semantic concerns), and Schrijver (1995: 55) offers a different etymology for the Celtic material (comparing instead OW *burr* 'fat, strong, thick', MIr. *buirre* 'swollenness'). None of this is compelling; and none of it accounts for *grossus*[1].

2.3 As an agricultural term, *grossus*[1] is an excellent candidate for Christol's theory (1.2). We may thus consider possibilities for a preform (pre-Lat.) **grōso-*, for which various PIE interpretations are conceivable, i. e. **grōso-* (or **ǵrōso-*), **groHso-* (or **ǵroHso-*), with voiced stop, and **g^{h}rōso-* (or **ǵ^{h}rōso-*), **g^{h}roHso-* (or **ǵ^{h}roHso-*), with voiced aspirate.[34] Most of this can be eliminated immediately: thus, there is no PIE **g^{h}res-* or **ǵ^{h}res-* (which might supply the basis for a lengthened *ō*-grade formation); there is no **greH-* or **ǵreH-* (which might supply the basis for *s*-formations **groH-s-o-* or **ǵroH-s-o-*); and for PIE **gres-* 'devour',[35] evidence for such a lengthened-grade formation is lacking. But an etymon for the laryngeal-final version with initial voiced aspirate is readily apparent, i. e. the root of Eng. *grow* (OE *grōwan*, OIc. *gróa*, etc. < **g^{h}roh$_{1}$-ye/o-*) and *green* (OS *grōni*, OHG *gruoni*, etc. < **g^{h}roh$_{1}$-ni-*).[36] Moreover, both *o*-grade and zero-grade *s*-forms are attested in Germanic, i. e.:

1. Gmc. **grōsō-*: MHG *gruose* 'sprout; sap', MLG *grōse* 'sap' (both f.);
2. Gmc. **grasa-* (with analogical onset): Go./OHG/OIc. *gras*, OE *græs* 'grass'.[37]

In addition, a zero-grade *s*-form may well be attested in Latin itself, in *grāmen* 'grass' (Cato+), in theory from pre-Lat. **grasmen* or **grā(s)men*. The first would be based on a root **gras-* 'devour' (as in Ved. *grásate*, Gk. γράω; cf. n. 35), an interpretation favored by Meillet (DELL: s. v.) and others (e. g. Watkins 2011: 33), in part based on assumptions about the Latin treatment of **g^{h}r-* (more on which

33 See e. g. WH: s. v.; IEW: 485 (ultimately from Osthoff and Fick, see WH for details).

34 I return briefly below to the problem of the development of **g^{h}r-* and **ǵ^{h}r-* in Latin.

35 Thus LIV²; according to a different conception, **gras-* (e. g. IEW: 404; Watkins 2011: 33).

36 IEW: 454; Watkins 2011: 32 (**ghrē-** < **g^{h}reh$_{1}$-*); EDPG: 191 s. v. **grōan-*.

37 On **grasa-* as a "secondary *s*-stem created to the verb *grōan-*" (and thus related to **grōsō-* as an ablaut variant), see EDPG: 187 (s. v. **grasa-*).

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below). But the Germanic comparanda certainly favor the alternative analysis via the 'grow' root, i. e. *$g^hr̥h_1$-*mn̥* or (more likely, with *s*-formant as in Germanic) *$g^hr̥h_1$-*s*-*mn̥* (Schrijver 1991: 487; EDL: s. v.; EDPG: s. v. **grasa*-). Thus Lat. *grossus*[1] (< pre-Lat. **grōso*-) may supply the "missing (Italic) link" in this Germanic/Italic network of agricultural terminology.

Given the nature of the evidence, it is difficult to determine exactly what kind of *s*-extension may be reflected in these forms. Suggestive of an *s*-enlargement (though hardly decisive) is the appearance of a dental enlargement in some related material, such as OE *græd* 'grass', MHG *graz* 'young conifer branch'. In either case — i. e., whether we have to do at bottom with an *s*-present or an *s*-enlargement — there is no difficulty with the assumption of a deverbal or deradical "τομός form" *g^hroh_1-*s*-*ó*- 'growing'; cf. 1.5 above and the forms pointing to **Ḱlow*-*s*-*o*- (Av. *sraoša*-, OCS *sluxŭ*, n. 30). It is possible, moreover, that despite the chronology of the attestations (2.1), the adjective *grossus* 'unripe' could be the older form, with *grossus* (m., f.) 'unripe fruit (esp. fig)' as an early substantivization (later also *grossum* n.); cf. *pōmus, pōmum* '(tree-)fruit' (Cato+) < **po*-(h_1)*em*-*o*-/**po*-(h_1)*om*-*o*- or **po*-h_1*m*-*o*- 'Abgenommenes'.[38]

Note finally that the above proposal intersects with — and may shed light on — the notorious problem of Latin historical phonology concerning the double reflex of word-initial *g^hR- (and *$ǵ^hR$-): sometimes *R*- (e. g. *rāvus* 'tawny, grey' < *$g^hr̥h_1$-*wo*-),[39] but sometimes *gR*-, as in *grossus*, according to this analysis, and possibly in *grāmen* (cf. above) and *glārea* (dissimilated from **grārea*) 'gravel' (Zair 2013). But this must be reserved for separate treatment elsewhere.

2.4 How to interpret "*grossus*[2]" remains unclear. One could speculate that this is originally the same word as "*grossus*[1]", but with a semantic split, along roughly the following lines:

1. *'growing; green' > 'unripe, immature (of fruit)' > 'immature fig' (and other fruit, cf. 2.1 (3.)), whence "*grossus*[1]" as an agricultural *terminus technicus*;
2. *'growing, ripening' > 'plump, fat' (cf. the background suggested in 1 for *crassus*), with a later development to 'coarse, rough, crude' (likely influenced by *crassus*, as often assumed), whence "*grossus*[2]" — perhaps originally a rustic or low-register usage alien to the standard language, which became naturalized in literary Latin only very late.

38 On this etymology of *pōmus/-um*, see EDL: s. v. Weiss 2010a: 229 n. 313.

39 Cf. virtual *g^hreh_1-*wo*- in Gmc. **grē-wo*- (OHG *grāo*, OE *grǣg* 'grey', etc.).

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3 Other forms

3.1 Some additional Latin material could be considered in terms of the above proposals, but most of it cannot (or should not) be pressed very far; e. g.:

1. There is little certainty about the background of Lat. *crīsāre* (occasionally *crissāre*, see TLL: s. v. *crīsō*) 'move the haunches suggestively (as in intercourse)' (Lucil.+),[40] for which various preforms have been proposed: see WH for **kreit-s-* or **krīt-s-*[41] and EDL for **kris-* or **krīs-*[42] (both s. v. *crīsō*). For the latter, as de Vaan notes, one should probably assume a pre-Lat. **krīss-*, with gemination, whence degeminated *crīsāre*. This pattern does not match that of *crassus* or *grossus*, and one must also reckon, in such a word, with the real possibility of expressive gemination. There is also the question, again, as to whether rhotacism in an original sequence /krīsV-/ could have been blocked by the onset cluster with /r/ (cf. 1.2 ad init.; more on this point below).
2. Lat. *vīsīre/vissīre* 'fart softly' (Lucil.+) may be related to *vēsīca* 'bladder' (Pl.+), and at least partly involves onomatopoeic effects (see EDL: s. v. *vēsīca*).

3.2 Other forms that seem worth pursuing at first glance, given their semantic proximity to *crassus* and *grossus*[2], turn out to be intractable:

1. Lat. *bassus* 'thick' (glosses only) and *Bassus* (Cic.+, 1.1)[43] is usually thought to have a Sabellic *b-*, given the Campanian profile of *Bassus, Bassius*, etc. (see DELL: s. v. *bassus*, TLL: s. v. *Bassus*). But even if such a "rustic" background is granted, there is no available etymology.
2. For Lat. *spissus* 'sluggish, dense, thick' (Naev.+), de Vaan (EDL: s. v.) observes that "[s]ince *spissus* is reminiscent of *crassus, grossus,* it may contain an expressive geminate — hence a possible preform would also be **spīsus*." His thinking was partly guided by the fact that the traditional etymology, based on an alleged **spid-to-*,[44] is far from compelling. But etyma that would support, for example, a PIE **spiH-s-o-* are not forthcoming. It is also not encouraging

40 With reference to women; the corresponding term for males is *cēvēre*.

41 IEW: 937, **(s)krei-t-* (to the putative basis 3. **(s)ker-* 'drehen, biegen', 935ff.): MIr. *crith* 'shivering, fever', OHG *scrit* 'Schritt', etc.

42 IEW: 937, **(s)krei-s-* (to the same root as above): Go. *us-hrisjan* 'shake out', Lat. *crīnis* 'hair (of the head)', etc.

43 With a development to 'low, short' (CGL 4.210.17, *Not. Tir.*+; Ital. *basso*, Fr. *bas*, etc.).

44 See EDG: s. v. σπίδιος for the standard comparanda from Greek, mainly with meanings like 'broad, extensive, distant'. Note that Hom. (gen. sg.) σπιδέος (Λ 754), of uncertain meaning, is unusable, despite Frisk (GEW: s. v. σπίδιος), Chantraine (DELG: s. v. σπιδέος), and others; cf. West 2006: 346 ad loc.: "vox ignota, fort[asse] nomen propr[ium]".

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that Lat. *spissus* is primarily attested in poetry, which reduces (though does not quite eliminate) the possibility of a "rustic" (or other lower-register) source.

There is, however, one final form that seems much more promising.

4 Lat. *classis* 'levy (of an army), class (of assembled people), fleet'

4.1 There is universal agreement that the earliest meaning of Lat. *classis* f. (*Lex Reg.*+) was '(roll-)call, appeal, summons', with reference to the levy of citizens called to arms (*classis iūniōrum* 'levy of youths'), whence terms for various types of levied troops: *classis clipeāta*,[45] *classis prōcincta*,[46] as well as troops serving at sea.[47] With the developing use of *exercitus* to refer to the land army, *classis* became specialized in the meaning 'fleet': cf. CIL: I², 25 (Columna Rostrata) c]lasesq(ue).navales, where the adjectival specifier is still necessary.[48]

The relationship with Lat. *calāre* 'call out, summon' (Varro+) — a verb almost exclusively attested, along with its derivatives, in archaic technical (especially religious and legal) terminology[49] — was clear to the Romans themselves (Quintil. 1.6.33: *classis a calando*) and is accepted by some modern authorities.[50] But the difficulty in accounting for the sequence *-ăss-* in *classis* is severe; indeed, Meillet (in DELL) opted for a desperate solution: "Terme technique qui peut être emprunté à l'étrusque".[51] Others (see WH: s. v.) have sought to account for the *-ss-* on the basis of the dental stem seen in Gk. κέλαδος 'noise, din' (whence an alleged **klād-ti-* or **klad-ti-*) and/or Ved. *krándati* 'cries out'; but this, too, is an extreme solution,

45 P.F. 48.22L: "classes clipeatas antiqui dixerunt, quos nunc exercitus vocamus".

46 P.F. 49.10L: "classis procincta, exercitus instructus" (similarly P.F. 251.19L, F. 294.3L, P.F. 295.2L).

47 P.F. 251.20L: "Vetustius enim fuit multitudinem hominum, quam navium, classem appellari".

48 In this text also claseis, on which see Gerschner 2002: 133f. Even if (as is often thought, following Wackernagel) the text is an Imperial concoction, its phraseology is clearly based on archaic material. Cf. also *classis* 'army, pl. troops' as an occasional archaism in Classical authors (e. g. V. *Aen.* 7.716 *Hortinae classes* 'the troops of Hortus'). On the semantic development of *classis*, see DELL: s. v. for additional details.

49 E. g. *calātor* '(ritual) crier' (already kalatorem in the Forum Inscription), *calendae/kalendae* 'beginning of the month' (Pl.+); additional material below.

50 IEW: 549; WH ["v(iel)l(eicht)"]; EDL: s. v. *classis*.

51 Cf. Weiss (2011: 316) on *classis* (similarly desperate, but in a different way): "etymology uncertain".

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and is rightly rejected by de Vaan (EDL: s. v.; similarly Schrijver 1991: 185). As de Vaan points out, the connection (especially with κέλαδος) is "semantically unattractive"; and in formal terms, any analysis based on a double-dental sequence is merely a "paper reconstruction", lacking support for the presumed double-dental morphological sequence itself. He notes further that **klād-ti-* would have developed to **klāssi-*, which should then have undergone geminate simplification to **clāsi-* (like *cāsus* < *cāssus*, etc.; 1.2).[52] It may be added that even κέλαδος cannot justify a short-vowel preform **klad-ti-*, which has no basis whatsoever; and any relationship with Ved. *krándati* "reste imprécise" (Bader 1983: 55 n. 86; cf. EWAia: 1, 408 s. v.).[53]

Under these circumstances, including the obvious attractiveness of maintaining the connection with *calāre*, it is appropriate to consider whether *classis* could derive from an earlier **klā-s-i-*, which escaped rhotacism and entered the standard language as *classi-*, in the manner proposed above (following Christol's theory) for *crassus* and (at least) *grossus*[1] (if not also *grossus*[2]). An initial obstacle would seem to be the question of "rusticity", which such a word appears, at first glance, to lack. Yet this may be an overly narrow characterization of the conditions that might justify the "borrowing" or adoption of an unrhotacized form by speakers of more advanced phonostyles. In this case, one could consider the possibility that the word in question belonged to the military "Fachsprache" (*sermo castrensis*),[54] and was borrowed into the standard language as a technical term from that restricted (but linguistically influential) group of speakers. From the beginning of serious scholarship on *sermo castrensis* (Kempf 1901) and continuing into more recent times,[55] it has become widely assumed that this speech form was in fact extremely close, in many respects, to some type of *sermo vulgaris* or *sermo plebeius*. Adams (1995; 2003) has sounded a cautionary note on this point, especially in his writings about the language of the Vindolanda Tablets: these documents are characterized by a high degree of complexity in terms of their linguistic background, which includes a cohort of relatively well-trained scribes, some of whom were well-versed

52 Nor, for that matter, is there any evidence in support of vowel length in Lat. *classis* (Allen 1978: 75).

53 Bader's own version **klHd-ti-* [sic] cannot work phonologically, and her comparison with Gk. κληδών 'rumor' and similar forms (Bader 1983: 36) is specious (cf. κλε(ϝ)ηδών [*Od.*]).

54 In addition to the military material sketched above, note also *classicum (canere)* '(sound) the military trumpet call' (Caes.+, but possibly much older).

55 See, for example, Mosci Sassi 1983 (updating Kempf), and still more recent studies like that of Petersmann 1992 and (in a sociolinguistic vein, emphasizing the importance of *sermo castrensis* as a force contributing to the "mentality" of the Roman soldier) Maxlajuk 2002. On Roman Fachsprachen more generally: de Meo 2005; Langslow 2005; Fögen 2011 (with further references).

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in literary norms and even favored archaizing orthography. Yet Adams himself has uncovered a number of important "subliterary" features in these and other texts associated with *sermo castrensis*;[56] and other similarly sophisticated analyses of Latin military language[57] continue to emphasize the importance of a "vulgar" component, among other features. It thus seems reasonable to suppose that Lat. *classis* could have had a sociolinguistic background more or less comparable to the other terms studied above, even though it would have belonged to a different linguistic subculture.

A more serious issue is the question of whether evidence can be found for related formations in **-s-*.

4.2 Within Latin itself, there is in fact material that could plausibly be interpreted on the basis of archaic *s*-formations:

1. As is clear from Varro and Macrobius,[58] the "*Cūria Calābra*" referred to a place on the Capitoline Hill and to meetings held there on the Kalends by the pontifices. In these passages, moreover, the name of this assembly is explicitly associated with the words *Kalendae* and *calāre*, the latter including ritual utterances by the pontifices containing the form *kalō*. Similarly, in a discussion of derivational relationships among related lexical items, Varro writes (*L.* 5.13) *nec Curia Calabra sine calatione potest aperiri*, "nor can the *Calabra Curia* 'Announcement Hall' be opened without the *calatio* 'proclamation'" (transl. Kent 1951: 15). Likewise, Servius (ad V. *Aen.* 8.654) uses the phrase *Cūria Calābra* and associates the word with *Kalendae*.[59] There is, in short, no reason to doubt the connection between Lat. *calābra* and *calāre* (as clear to Varro and Servius as it is to modern etymologists: see WH; DELL; EDL: s. v. *calō*); and the form is clearly adjectival, in the fixed expression *Cūria Calābra*. Yet the standard interpretation in terms of the instrument suffix **-d^hro-*/**-d^hlo-* (so e. g. Leumann 1977: 314) is extremely unattractive, just as it is for Lat. *crēber*, as we have seen (1.5); indeed, these are the only two adjectival forms with this suffixation cited by Leumann. There is, to be sure, limited evidence that early Latin forms in *-c(u)lo-* and (as a result of liquid dissimilation) *-cro-* (based on the instrument suffix **-tlo-*, and otherwise in use as instrument

56 See in particular Adams 1977 (Claudius Terentianus); 1994 and 1999 (Bu Njem ostraca and poetry).

57 A fairly recent example is Pérez Castro 2005.

58 Varro, *L.* 6.27; Macrobius, *Sat.* 1.15.10; cf. also *Fasti ann. Iul. Praen. Ian. 1* (CIL: I², p. 231) […IN.CAPI]TOLIO.IN.CVRIA.CALA[BRA …].

59 See TLL (s. v. *curia Calabra*) for other traces of such a tradition.

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nouns) were occasionally pressed into service in adjectival usage.[60] But there is no evidence at all for any "instrumental" sense for either *crēber* or *calābra*, both of which are solely attested as adjectives. Thus, just as *crēber* is better interpreted in terms of a (pre-Lat.) **krē-s-ro-*, it seems legitimate to posit a pre-Lat. basis **klā-s-ro-* for *calābra*, with the further assumption (an easy one) that the resulting **clābro-* was remodeled to *calābro-* after *calāre* (and/or other derivatives with which it was transparently associated, such as *calātiō*, as in Varro's etymological comment).[61]

2. The well-attested gloss form *conclassāre*, placed with *classis* by both WH and DELL (though omitted in EDL), may be considerably more interesting than it has seemed hitherto. For one thing, it is assigned two quite different meanings, of which the most frequent is "classem iungere" (and similar variants).[62] But this has the look of a mere etymological guess, based on the productive comitative sense of Lat. *con-* grafted onto Class. Lat. *classis*, and perhaps influenced by Classical attestations of similar phraseology, e. g. *adiungit classem et exercitum* (Cic. *Att.* 4.1.7.6). More isolated, and with an opaque meaning from the point of view of Classical Latin, is the gloss *conclassare* "conclamare" (CGL 5.593.51), which, in view of its semantic closeness to the actual etymological basis of *classis* (cf. above), may reasonably be supposed to preserve something ancient. It is this form, moreover, which is also likely to have survived into Romance,[63] and which may thus have a reasonable claim to subliterary status.[64] Yet such a form cannot be derived from the *i*-stem noun *classis* itself, which should have produced a denominative [x]*classīre*.[65] The denominative pattern

60 See Leumann (1977: 314) on secondarily adjectival *rīdiculus* 'absurd' (Pl.+), *ēluācrus* (*lābrum ēluācrum* 'basin for rinsing-out', Cato *Ag.* 10.4, 11.3; cf. *lavācrum* 'tub' [Gell.+]) and *lūdicrus* (*artem lūdicram* Pl. *Aul.* 626, vs. *lūdicrum* 'plaything, sport', e. g. Cat. 61.24); recent discussion of such secondary adjectives (with additional examples) in Eichner 2013: 29f. The testimony of a form like *ēluācrum* may also be weakened if it fundamentally reflects an appositional usage; so e. g. OLD: s. v.

61 Lat. *celeber* 'famed, frequent(ed)' is often placed with *calāre* (cf. EDL: s. v. with reference to Bader 1983: 43f.); and if this is correct, it could well involve an *s*-formation. But the etymology and formation of the word are too unclear to allow it to be used as evidence; see Nussbaum 1999: 388f. with 411 n. 67.

62 Thus "classem iungere" at CGL 5.57.9, vs. "adiungere classem" (4.222.52, here spelled *conclasare*; a trace of archaic orthography?), "coniungere classes" (5.596.21), etc.

63 Cf. OFr./OProv. *clas* 'noise, bell-ringing', Ital. *chiasso* 'clamor, ruckus, annoying noise', among other forms.

64 The Romance data are somewhat complex, and cannot be explored here; see, however, REW: 186 (#1965, **classum*), 200 (#2115a, **conclassāre*); DEI: 2, 894f. (s. v. *chiasso*[1]); DELI: 1, 230 (s. v. *chiàsso*).

65 See the discussion of *i*-stem denominatives by Mignot (1969: 64f.).

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with suppression of *-i-* for *i*-stem adjectives (*gravāre* ← *gravis* 'heavy', *levāre* ← *levis* 'light', etc.; Leumann 1977: 546) is not relevant, as there is no reason to think that *classi-* was ever adjectival. For a denominative interpretation, then, the unmarked assumption would be that *-classāre* should be based on either an athematic nominal stem **klā-s-* or a thematic **klā-s-o-*. Alternatively, given the prefix and 1st-conjugation form, we might have an "*ā*-intensive" of the type (*sternere* 'spread' →) *cōnsternāre* 'confound', (*pellere* 'push' →) *compellāre* 'call upon', etc. (Leumann 1977: 549f.) or a "deverbal *ā*-verb" (de Vaan 2012), implying a thematic **classere*, which might have a basis in an *s*-present **klā-s-e/o-* (cf. 1.5 on **crassor*).

3. The adjective *clārus* 'loud' and (secondarily) 'clear, bright, famous' (Naev.+) is generally assumed to continue a simple *ro*-adjective **$k_{\text{l}}^{\text{u}}h_1$-ró-* to the root of *calāre*, and this is indeed the simplest assumption.[66] But in theory, one could posit a thematic (possessive) derivative of an original *s*-stem verbal noun (roughly 'calling out, shouting', whence conceivably 'loudness'): thus (with the zero-grade format seen above for *russus* and suggested as a possibility for *crassus*) **$k_{\text{l}}^{\text{u}}h_1$-s-o-* 'having loudness' > pre-Lat. **klā-s-o-* 'loud'. Alternatively, as with *crassus*, the same formal and semantic result could be achieved by a "zero-grade τομός form" based on an *s*-extended root (**$k_{\text{l}}^{\text{u}}h_1$-s-o-* 'calling out' > pre-Lat. **klā-s-o-* 'loud').

One or the other of these theoretical possibilities may be more attractive for the archaic ritual term (with reference to the Fetiales) *clārigāre* 'formally demand satisfaction (from another state, in a ceremonial declaration of war)' (Plin. *Nat.* 22.5), with earlier-attested verbal noun *clārigātiō* 'reparation, redress' (Livy+), and (according to an attractive conjecture of Lachmann) a frequentative *clārigitat* in Lucretius (5.947).[67] It has usually been assumed[68] that the basis of the first member is the adjective *clārus*, with the formation in *-igāre* modeled on *pūrigāre* 'cleanse' (Pl.+, Class. Lat. generally *purgāre*), this in turn being based on the adjective *pūrus* 'clean, pure'. It now seems probable, however, that OLat. *pūrigāre* is based not on *pūrus*, but rather on a noun **pūr* 'fire', comparable to U. **pir** 'id.'.[69] Thus, if this account of *pūr(i)gāre* is correct, *clārigāre* can be evaluated on its own, in comparison not only with *pūrigāre*, but with other archaic formations of this kind,

66 Cf. **$k_{\text{l}}^{\text{u}}h_1$-mo-* → denominative *clāmāre* 'shout'.

67 On the Lucretius usage and, more generally, the word's antiquity as part of the "vocabulaire juridico-rituel des féciaux", see Mignot 1969: 343, with further references. This clearly archaic material is (regrettably) omitted in EDL (s. v. *clārus*).

68 E. g. Mignot 1969: 340; Leumann 1977: 550.

69 Thus EDL: s. v. *pūr(i)gō* (following Dunkel 2000: 94); Weiss 2011: 403 n. 8.

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in particular *nāvigāre* 'go by ship' (Pl.+), *iūr(i)gāre* 'quarrel' (Pl.+), and *lītigāre* 'bring a dispute' (XII Tab., Pl.+). As made clear by Dunkel (2000: 94, developing Forssman's analysis of *nāvigāre*), at least the first two involve archaic compounds built with athematic first members and second members based on the zero grade of $*h_2eǵ$-, with basic meaning 'drive, lead', but also (in Latin juridical language) 'bring': thus virtual $*neh_2u$-$h_2ǵ$-*(o-)* 'driving the ship' and $*yewos$-$h_2ǵ$-*o-* 'bringing the oath', whence (in part via analogical developments) pre-Lat. **nāw-ago-* and **yowz-ago-*.[70] One could thus postulate, for what appears to be the highly archaic legal/ritual term *clārigō*,[71] an original compound of the same sort, i. e. (pre-Lat.) **klās-ago-* 'bringing the summons' (vel sim.), with first member based either on an archaic *s*-stem (as in **yewos* and *iūrigāre*), or else a "virtual root noun", based on a **klās-* that in PIE terms would have been the weak stem $*kl̥h_1$-*s-* of an *s*-present.[72] Note finally that this analysis of the *clāri-* of *clārigāre* in terms of an earlier **klās-* or **klāsi-* is independent of the question of whether or not *clārus* itself reflects a related *s*-form, as opposed to an *s*-less *ro*-adjective.

4.3 The above hypotheses would be strengthened by external evidence for an *s*-formation, and this can in fact be found. It is generally acknowledged that Lat. *calāre* (along with the *ē*-verb in U. **kařetu**, **kařitu**/*carsitu* 'he shall call') belongs with Gk. καλέω 'call' and Hitt. *kalliš*-^{zi} (also *kalleš*-^{zi}) 'call, summon, invite (to a ritual meal)', despite disagreement about the exact form of the root itself.[73] The Hittite formation is sometimes explained as a stative in $*$-*eh$_1$-s-* (thus Kimball 1999:

70 A somewhat different account for the second one is offered by Hackstein (2012: 90), who suggests that *iūrigāre* could have been formed directly from the verb phrase *iūre agere*, on the analogy of *nāvem agere* and *nāvigāre*, and without an intermediate stage such as **iūrigus* or the like. The analysis of *lītigāre* is more complex. Joseph (1986: 123) argues for an *i*-stem *lī-ti-* (as opposed to a consonant-stem *lī-t-*); but the case is not secure. If Nussbaum is correct in assuming that *-ti-* after stressed long vowel does not syncopate in Latin nom. sg. forms (2004: §1.1.1.5, cf. *vītis, crātis*), the nom. sg. *līs* (which Joseph does not mention) points rather to a plain *t*-stem, as apparently assumed by Gerschner (2002: 105). If so, pre-Lat. **(s)līt-ago-* or **(st)līt-ago-* can be a form of the same type as **yowz-ago-* (in the same semantic field); but if *lī-ti-* is preferred, it is not difficult to account for *lītigāre* secondarily, on which see Dunkel 2000: 94.

71 Predictably with rhotacism, for this high-style lexical item, vs. *classis* (from *sermo castrensis*).

72 Cf., once again, PIE **ḱlew-s-*, with the element **ḱlews-*/**ḱlus-* treated as a root in formations like Ved. *śrúṣti-* 'obedience'; within Latin, cf. also *fās* 'divine law', if this belongs (as $*d^heh_1$-*s-*) with an *s*-extended version of $*d^heh_1$- (DELL: s. v.; Watkins 2011: 18). If one follows Hackstein's account of *iūrigāre* (n. 70), one could also think in terms of an original verb phrase, such as **klās agere* or even (with *i*-stem) **klāsim agere*, without the need for an intermediate compound of the sort **klās-ago-*.

73 Thus variously $*kleh_1$- (LIV² and others, e. g. Meiser 2003: 99 n. 15, Zair 2012: 91), $*kelh_1$- (Schrijver 1991: 185, EDL: s. v. *calō*), $*kalh_1$- (Oettinger 1979: 185, 197; Weiss 2011: 41, 82), $*kelh_2$-/$*kleh_2$-

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412), based in part on its appearance as *kalleš-* (actually the minority spelling), and in comparison especially with the Umbrian *ē*-verb. But this analysis is to be rejected. The semantics are not favorable for a stative, either in Hittite or Umbrian; and for its part, the Umbrian form is just as likely to be a zero-grade **-eye-* formation,[74] which could easily be iterative.[75] Kloekhorst has argued persuasively (EDHIL: s. v. *kalliš-*^{zi}; Kloekhorst 2009: 246f.) that the Hittite verb should instead be analyzed as an athematic *s*-present, in his view with original ablaut **kélh$_1$-s-ti*/**klh$_1$-s-énti*. One can question some of his assumptions about the behavior of anaptyxis (and its reflection in the Hittite spellings),[76] as well as his evaluation of the significance of the spelling *ga-li-iš-ša-an-zi* (1×, NS); but the overall morphological interpretation of the Hittite material in terms of an ablauting *s*-present of some kind is very attractive, and seems likely to be correct. Less appealing, but perhaps marginally possible, is the alternative theory of Oettinger (1979: 185, 197), according to which the Hittite verb was back-formed from a noun **kalleštar* 'invitation' (inferable as the basis for the attested noun *kallištarwana-* 'feast'), based on a primary *s*-stem **kálh$_1$-es-*, which Oettinger also sees as the source of Gk. καλέω. Apart from the analyses just proposed for some isolated Latin material (4.2), there is no independent evidence for a primary *s*-stem (especially in Greek, where καλέω is not likely to have such a source), and no reason to think that the Hittite verb was formed in this way.[77]

4.4 The isolation of Lat. *classis* makes it difficult to decide on the form's morphological background, for which the following two options seem thinkable, one "denominal" and the other "deverbal":

1. If there existed a thematic **klā-s-o-* ('having loudness' or 'calling out'; 4.2 (3.)) as the basis for *clārus*, or else a thematic **klā-s-o-* (with similar meaning) underlying *conclassāre* (if the form is denominative; 4.2 (2.)), a regular

(Watkins 2011: 40), **kleh$_1$-* for καλέω and the Umbrian material vs. **kleh$_2$-* for Lat. *calāre* (Untermann 2000: 360).

74 Thus LIV² and others, e. g. Kölligan 2002: 154 (similarly for Gk. καλέω, on which see already Harðarson 1993: 82 n. 98); Meiser 2003: 67.

75 See Kölligan 2002 for the semantics of verbs of this type. According to another view (Schrijver 1991: 400), the *ē*-inflection is a purely Umbrian development of the weak stem-form **kale-* arising from a root athematic 3 pl. **klh$_1$-énti*. On the morphological analysis of Lat. *calāre* beside the Umbrian *ē*-verb, see most recently de Vaan 2012: 320.

76 For some recent discussion of anaptyxis processes in Hittite, see Melchert 2013: 178ff.

77 Kloekhorst (EDHIL: s. v. *kalliš-*^{zi}) judges this theory to be "improbable"; as he points out, the noun **kallištar-* can easily reflect a deverbative **kelh$_1$-s-tr-*. The possibility that Gk. καλιστρέω 'καλέω' (Demosth.+) may have its source in Asia Minor (Oettinger 1979: 197 n. 37) is intriguing, but offers no support for this theory.

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i-substantivization[78] would produce a feminine abstract **klās-i-* 'summons, levy'; such a form, borrowed from *sermo castrensis* into the standard language (and assuming a phonological development of the type proposed by Christol) would yield Lat. *classis*.

2. Alternatively, beside an athematic *s*-present stem **kl̥h₁-s-* (or an *s*-enlarged form of the same shape), one can posit the existence of an *i*-stem verbal noun **kl̥h₁-s-i-* 'calling out' (> **klās-i-* > *classis*/?*clāri-*): cf. the primary *i*-stem action nouns with zero grade of the root seen in Ved. *dṛśi-* 'sight', *yudhi-* 'fighting', etc., underlying the Vedic dative infinitives *dṛśáye* 'to see', *yudháye* 'to fight', etc., a formation seen also in some isolated feminine action nouns (e. g. RV *kṛṣí-* 'tillage', AV *rúci-* 'gleam'), beside formations with other vocalisms.[79] Similar zero grade forms are attested elsewhere, as with some archaic material in Greek,[80] beside formations with other root vocalisms in Greek and elsewhere.[81] This "direct to *i*-stem" deverbal account is derivationally simpler than the first, as it lacks the intermediate substantivization, and would find support in a Hittite *s*-present. But an immediate objection would seem to be the fact that such *i*-stem verbal nouns tend to appear in Latin in the category of nouns with nom. sg. *-ēs* (often thought to derive ultimately from hysterokinetic *i*-stems):[82] thus *caedēs* 'murder' (: *caedere* 'strike, slay'), or (with an isolated zero grade) *clādēs* 'destruction' (: *-cellere* 'strike'), etc. Nevertheless, plain *i*-stem verbal nouns with nom. sg. *-is* are also attested, of which the best example may be *scobis* f. 'sawdust, shavings' (Varro+),[83] with concretized meaning ('scrapings') from *'scraping' (: *scabere* 'scrape'). Clearly archaic, but without attested verb in Italic, is Lat. *scrobis* 'hole, pit' (Pl.+), with verbal comparanda in Balto-Slavic, Germanic, and Celtic (LIV²: s. v. **(s)kreb-*). It is thus perfectly

78 The pattern is by now very familiar; see Weiss 2011: 314f. and 2013: 346ff. for Latin and Celtic examples, with further references. (Cf. also n. 22 above on **h₁rud^{h}-s-o-* → **h₁rud^{h}-s-i-*.)

79 E. g. RV *rájі-* 'direction', *vyáthi-* 'path', etc.

80 E. g. the *i*-stem **tr̥k^{w}-i-* 'turning' (cf. Lat. *torquēre* 'twist') underlying Myc. *to-qi-de* 'spiral' and related Mycenaean forms, or the frozen acc. sg. **k^{w}l̥h₁-i-* 'id.' in πάλιν (adv.) 'backwards' (to **k^{w}elh₁-* 'turn': Hom. [Aeol.] πέλομαι etc.).

81 Thus *ē*-grade in δῆρις 'strife' (Hom.+, to δέρω 'flay'), cf. OCS *rěčĭ* 'speech' (: *rekǫ* 'speak'), etc. Garnier's attempt (2014: 150ff.) to interpret the *ē*-grade of δῆρις as the result of the "Kortlandt effect" produced by a primordial reduplication (**dé-dr(H)-i-* > **dé-dr-i-* > **dé-h₁r-i-*) is unconvincing, since there is no reason to expect that the phonological process in question would apply to sequences with such a syllable structure. For surveys and discussion of *i*-stem *nomina actionis*, see Barschel (s. d.): 148ff.; Grestenberger 2009: 13ff. and 2014; Rau 2009: 181; Stüber 2012: 131.

82 See e. g. Klingenschmitt 1992: 114ff.; Weiss 2011: 243f. (with 243 n. 13).

83 The antiquity of the *i*-stem is supported by *scobīna* 'scraper, rasp' (Pl. *fr.* 91 [Lindsay] = 93 [Monda] +), which may be deinstrumental (**skób-i-h₁* 'with scraping' + **-no-*).

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plausible (and indeed part of the potential morphological interest of the word) that an isolated form like *classis* could continue an archaic verbal noun of the type Ved. *dṛśi-* etc., although questions remain about the original inflection (and the morphological interpretation) of such forms. Thus according to Rau (2009: 181), deverbative *i*-stem substantives were originally acrostatic (with both the *o/e-* and *ē/e*-types represented in principle); nevertheless, "the type has been remodeled with proterokinetic inflection in Vedic" (2009: 181 n. 152), hence the zero-grade forms of the type *rúci-* and the like. Thus the zero grade in **kl̥h₁-s-i-*[84] could be explained in various ways. We might have a "proterokinetic remodeling" of the type just mentioned; or (Nussbaum ap. Rau 2009: 177 n. 143) we might have an internally-derived proterokinetic derivative (originally adjectival) of an acrostatic *i*-stem substantive;[85] or we might have a substantivization of an original *o*-stem adjective, such as a "zero-grade τομός form" (cf. 1.5 [on *crassus*] and 4.2 (3.) [on *clārus*] above, and Rau 2009: 181 n. 153 on *rúci-* beside *rucá-* and *rocá-*, et sim.). These indeterminacies can only be signaled here.

4.5 A final observation concerns the question asked near the outset (1.2, ad init.) about the failure of rhotacism to apply in a sequence like **krāso-* (and thus also **grōso-* [2] and conceivably **krīsV-* [3.1 (1.)]). If Lat. *classis* (which lacks a rhotic altogether) has an explanation like that proposed above, it would support the idea that — as distinct from rhotacism exceptions like *miser*, *caesariēs*, *rosa* etc. — the failure of rhotacism in forms like **krāso-* and **grōso-* need have nothing to do with the presence of an onset cluster /Cr-/, especially if Roberts (2012: see n. 6 above) is correct in analyzing the *miser*-type exceptions in terms of a constraint against **rV̆r* sequences, as opposed to licit /rV̄r/ (e. g. *rōs* 'dew', gen. *rōris* etc.).

5 Summary of conclusions

The main conclusions arrived at above can be summarized as follows:

1. according to Christol's theory about *s*-gemination in forms that have failed to undergo rhotacism (1.2), Lat. *crassus* 'thick etc.' and *grossus* 'immature (of fruit); immature fig' (both without clear etymology) could go back to pre-Lat. **krāso-* and **grōso-*;

84 And likewise in similar Greek forms like those in n. 80 above.

85 This, however, would involve an undesirable re-substantivization: 'calling out' [*o/e* or *ē/e* noun] → 'calling out' [*e/Ø* adj.] → 'summoning' [f. *i*-abstr.].

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2. etyma for both (and possibly also for Late Lat. *grossus* 'thick, fat') may therefore be found in *s*-extended forms of PIE roots that basically mean 'grow' (cf. Lat. *crēscere, Cerēs*, OHG *hirso,* etc. for *crassus*, and Lat. *grāmen*, Gmc. **grasa-* and **grōsō-* for *grossus*);
3. Lat. **crassundo-* (→ *crassundia* [1.5]) may support the idea that Lat. *crēscere* ultimately goes back to an old *s*-present;
4. the hitherto unexplained form of Lat. *classis* (originally 'levy, summons', cf. *calāre* 'call out, summon') can be accounted for in the same way as *crassus* and *grossus*, given probable evidence for an *s*-present in Hitt. *kalliš-*^{zi} 'call, summon' and possible evidence for *s*-based forms within Latin itself, i. e. *calābra* (in *Cūria Calābra*), *conclassāre*, and *clārigāre*.

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Abbreviations

AiG	Jakob Wackernagel & Albert Debrunner (1930–1957). *Altindische Grammatik*. 2nd ed. 3 vols. Göttingen: Vandenhoeck & Ruprecht.
CIL	Eugenius Bormann et al., eds. (1876–1980). *Corpus Inscriptionum Latinarum*. Berolini: Reimerum & De Gruyter.
DEI	Carlo Battisti & Giovanni Alessio (1966). *Dizionario etimologico italiano, II*. Firenze: Barbèra.
DELG	Pierre Chantraine (1999). *Dictionnaire étymologique de la langue grecque. Histoire des mots*. 2nd ed. Paris: Klincksieck.
DELI	Manlio Cortelazzo & Paolo Zolli (1979). *Dizionario etimologico della lingua italiana, I*. Bologna: Zanichelli.
DELL	Alfred Ernout & Antoine Meillet (1985). *Dictionnaire étymologique de la langue latine. Histoire des mots*. 4th ed. Paris: Klincksieck.
EDG	Robert S. P. Beekes (2010). *Etymological Dictionary of Greek*. With the assistance of Lucien van Beek. 2 vols. Leiden & Boston: Brill.
EDHIL	Alwin Kloekhorst (2008). *Etymological Dictionary of the Hittite Inherited Lexicon*. Leiden & Boston: Brill.
EDL	Michiel A. C. de Vaan (2008). *Etymological Dictionary of Latin and the other Italic Languages*. Leiden & Boston: Brill.

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EDPC	Ranko Matasović (2009). *Etymological Dictionary of Proto-Celtic*. Leiden & Boston: Brill.
EDPG	Guus Kroonen (2013). *Etymological Dictionary of Proto-Germanic*. Leiden & Boston: Brill.
EWAia	Manfred Mayrhofer (1986–2001). *Etymologisches Wörterbuch des Altindoarischen*. 3 vols. Heidelberg: Winter.
GEW	Hjalmar Frisk (1960–1972). *Griechisches etymologisches Wörterbuch*. 3 vols. Heidelberg: Winter.
IEW	Julius Pokorny (1959–1969). *Indogermanisches etymologisches Wörterbuch*. Bern: Francke.
LEIA	Joseph Vendryes & Pierre-Yves Lambert (1959–). *Lexique étymologique de l'irlandais ancien. Lettres A, B, C, D, M, N, O, P, R, S, T, U*. Paris & Dublin: Centre National de la Recherche Scientifique & Dublin Institute for Advanced Studies.
LIV²	Helmut Rix (2001). *Lexikon der indogermanischen Verben. Die Wurzeln und ihre Primärstammbildungen*. Unter Leitung von Helmut Rix bearbeitet von Martin J. Kümmel, Thomas Zehnder, Reiner Lipp, Brigitte Schirmer. 2nd ed. Wiesbaden: Reichert.
LIV² Add.	*Addenda und Corrigenda zu LIV²* (n.d.). URL: http://www.martinkuemmel.de/liv2add.html.
OLD	Peter G. W. Glare, ed. (1982). *Oxford Latin Dictionary*. Oxford: Oxford University Press.
REW	Wilhelm Meyer-Lübke (1935). *Romanisches etymologisches Wörterbuch*. 6th ed. Heidelberg: Winter.
TLL	*Thesaurus linguae latinae* (1900–). Stuttgart, Leipzig et al.: Teubner, de Gruyter.
WH	Anton Walde & Johann B. Hofmann (1938–1956). *Lateinisches etymologisches Wörterbuch*. 2 vols. Heidelberg: Winter.

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